MSc. in Computing
Practicum Approval Form

Section 1: Student Details

Project Title:	Secure Data Integrity and Anomaly Detection Framework for Corporate Banking Systems
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Chosen major:	Secure Software Engineering
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Date of Submission	21 November 2025

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

The practicum is aimed at creating a two-layer security system concerning Corporate and Investment Banking (CIB) systems that combines the data integrity verification with AI-based anomaly detection into one united system. Rather than the integrity checks and anomaly detection being independent or separate modules, the project has a secure pipeline where they first perform the validity of all financial records through hashing algorithms, and then they are run through the statistical and machine-learning model. The combination of these methods guarantees security of CIB workflows against both manipulated information and deviant behavioural patterns, which is a significant cybersecurity vulnerability in the current systems.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. **Comparative analysis of machine learning algorithms for money laundering detection** — *Discover Artificial Intelligence (2025)*. *Sunday Adeola Ajagbe, Simphiwe Majola and Pragasen Mudali.* https://link.springer.com/article/10.1007/s44163-025-00397-4
2. **Transparency and Privacy: The Role of Explainable AI and Federated Learning in Financial Fraud Detection** — *Awosika, Shukla & Pranggono (2023)*. https://arxiv.org/pdf/2312.13334
3. **Online transaction fraud detection in the banking sector using machine learning techniques** — *Edelweiss Applied Science and Technology (2024)*. *S.*

Vasudevan, Vediyappan Govindan, Haewon Byeon. https://learning-gate.com/index.php/2576-8484/article/view/1781/632

4. **Fraud Detection in Banking: A Deep Learning Approach with Explainable AI** — *Koppireddy & Devi (2025)*. https://irojournals.com/jscp/article/pdf/7/3/4

5. **Real-Time Data Analytics for Fraud Detection in Investment Banking Using AI and Machine Learning** — *Munivel Devan , Chandrashekar Althati , Jegatheeswari Perumalsamy (2023)*. https://thesciencebrigade.com/cndr/article/view/267

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Available literature on financial fraud detection has relied mainly on applying machine learning to detect unfavorable behaviour patterns. As an example, Ajagbe et al. (2025) compare money-laundering detectors based on ML algorithms, and Awosika et al. (2023) study explainable AI and federated learning in privacy-preserving fraud analytics. On the same note, Vasudevan et al. (2024) assess online transaction fraud detection methods based on ML, and Koppireddy et al. (2025) assess fraudulent retail transactions detection using online deep learning with explainability. The contribution of Devan et al. (2023) to these ideas is based on investment-banking data streams but continues to apply mostly behavioural anomaly analysis.

In all these works, the general assumption is that input data is correct, complete and untampered. Before executing ML models, they do not ensure data integrity, identify silent changes or authenticate workflow. The proposal builds upon this literature by proposing a two-level security model, first with a tamper-free data integrity through the application of hashing algorithms, and second with an application of AI-based anomaly detection.

Moreover, Corporate and Investment Banking (CIB) presents its own challenges in contrast with generic fraud detection: low volume/high value transactions, multi-step approvals, manual interventions, and internal handoffs as well as insider driven tampering risks which are still left to be addressed and solved.

What are the research questions that you will attempt to answer? (200 words)

1. How can a secure data-processing pipeline be designed for CIB systems to ensure integrity, traceability, and resistance to tampering across end-to-end transaction flows? (Risk related to Data Integrity)

2. How effectively can machine-learning–based anomaly detection models identify synthetically injected behavioural anomalies in integrity-verified Corporate and Investment Banking (CIB) datasets? (Risk related to Anomalous behaviors)

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?

The practicum will be developed entirely in **Java 17+**, using **Maven** for dependency management. Development will take place in IntelliJ IDEA, with Gitlab for version control. Java libraries such as **WEKA** (for machine learning) because it integrates better with the Java environment and allows to develop lightweight, local ML experimentation without heavy dependencies, **Apache Commons CSV** (for dataset ingestion), and **BouncyCastle** (for cryptographic hashing/HMAC) will support the implementation.

- What coding/development will you do?

- A **Data Loader module** to parse CIB datasets into Java objects.
- An **Integrity Verification module** implementing hashing, HMAC, and hash-chain verification to detect tampering or replay attacks.
- An **Anomaly Detection module** using WEKA algorithms (Isolation Forest, Random Forest, clustering).
- A **Tampering Simulator** to insert controlled corruptions (timestamp shifts, record overrides, duplicate/replay injection).
- A **Results and Evaluation module** generating logs, alerts, and charts.

- What data will be used for your investigations?

We will use synthetic or inspired mock data for the practicum.

- How will it be generated?

  – The synthetic dataset will be programmatically generated in Java, creating realistic CIB-style loan and transaction records with correct sequences, timestamps, and approval workflows. It will be generated using sequence generators, time-series modelling, rule-based tampering simulation (for e.g., field swaps, timestamp modifications)

- Which behaviours/fraud patterns will be simulated?

  – The dataset will simulate both integrity violations (tampering, replay attacks, broken sequences) and behavioural anomalies (outlier loan amounts, irregular approvals, unusual repayment patterns) to test the full dual-layer security pipeline.

- How does synthetic data pair with the integrity framework?

  – Synthetic records will include fields specifically designed for integrity checks—hashes, previous hashes, versioning, and workflow steps—allowing the framework to detect tampering before ML models analyze behavioural anomalies.

- Is this data currently available, if not, where will it come from?

Yes, but we will create a realistic synthetic dataset for CIB workflows, we will take structural and behavioural inspiration by borrowing structure, field types, value ranges, and behavioural patterns from real-world corporate transaction datasets (MBD, PaySim), corporate loan datasets (SME Credit), ensuring that the mock data reflects realistic CIB operations while remaining safe and fully synthetic.

- What experiments do you expect to run?

- **Data integrity checks** using hashes, sequence validation, and field consistency to detect tampered loan or transaction records.
- **Anomaly detection** experiments will use statistical outliers, rule-based thresholds, and lightweight ML models (Isolation Forest/Random Forest) to identify unusual patterns.

The system will output flagged records with anomaly scores or integrity violations, demonstrating its effectiveness.

- What output do you expect to gather?

- Alerts showing **which records were flagged as tampered**.
- Anomaly scores for each transaction or loan record.
- Confusion matrices, precision/recall, F1 scores from WEKA models.
- Hash-chain breakpoints illustrating exact locations of data corruption.
- Charts (PDF/PNG) demonstrating anomaly spikes or integrity failures.

- How will the results be evaluated?

The results will be evaluated and considered by comparing flagged records against **manually introduced anomalies and tampered data** in the datasets or sample CSVs. Metrics such as **accuracy, precision, recall**, and the number of correctly identified integrity violations or anomalies will be measured. Successful detection of all injected irregularities will demonstrate the PoC's effectiveness.